Software Aim, Description, User Levels and General System Requirements

E-Commerce Platform

1. Project Topic

Smart Online Shopping System (SOSS) – An Intelligent E-Commerce Platform

The project focuses on designing a modern e-commerce platform that enables customers to browse products, compare options, place orders, and track deliveries through an integrated digital environment.

The system combines traditional online shopping capabilities with **advanced features such as intelligent product recommendations, predictive inventory analytics, group buying functionality, and sustainability tracking of delivery options**.

The platform aims to improve the online shopping experience by making product discovery easier, improving order transparency, and enabling sellers and administrators to manage products, inventory, and sales efficiently.

2. Description of the Project Topic

Aim of the System

The aim of the **Smart Online Shopping System (SOSS)** is to provide a centralized digital platform where customers, sellers, and administrators can interact within an integrated online marketplace.

The system is designed to support the entire online shopping lifecycle, including:

- product discovery
- product comparison
- order placement
- payment processing
- delivery tracking
- product reviews
- inventory monitoring
- sales analytics

In addition to core e-commerce functionality, the system incorporates **intelligent and innovative features** to enhance the user experience and platform efficiency.

These features include:

- personalized product recommendation systems
- predictive analytics for inventory management
- AI-based price optimization tools for sellers
- social commerce features such as group buying discounts
- sustainability tracking through carbon footprint reporting of delivery options
- video-based product reviews and unboxing content

The system therefore aims to create a **smart, transparent, and user-friendly online shopping ecosystem** that benefits both customers and product sellers.

3. User Levels

The Smart Online Shopping System supports **three main user levels**, each with different roles and permissions within the platform.

3.1 Customers

Customers are end users who browse products and purchase items through the platform.

Customers can perform activities such as:

- searching products using keywords and filters
- browsing products by category and subcategory
- comparing products based on price, specifications, ratings, and availability
- adding products to the shopping cart
- placing orders and completing purchases
- tracking the delivery of their orders
- submitting reviews and ratings for purchased products
- receiving notifications about order updates or price changes
- viewing personalized product recommendations

Customers can also participate in **group buying activities**, allowing multiple users to collaborate in order to unlock discounts.

Additionally, customers may access **video-based product reviews and unboxing demonstrations** to better evaluate products before purchasing.

3.2 Sellers / Vendors

Sellers are users who list products on the platform and manage their inventory and sales.

Sellers can perform activities such as:

- creating product listings with descriptions, images, and pricing
- updating product information and availability
- managing inventory levels
- monitoring sales performance and product analytics

- creating bundled product offers
- adjusting product prices using AI price optimization tools
- analyzing predictive inventory analytics for restocking decisions

These capabilities allow sellers to manage their online storefront efficiently and respond to market demand.

3.3 Administrators

Administrators oversee the overall operation of the platform.

Their responsibilities include managing the marketplace environment and ensuring system stability.

Administrators can perform tasks such as:

- monitoring platform orders and inventory
- managing promotional campaigns and discount codes
- supervising customer accounts and seller activities
- analyzing platform-wide sales reports and performance metrics
- ensuring the proper functioning of system services and integrations

Administrators play a key role in maintaining platform governance and operational integrity.

4. General Requirements to be Fulfilled by the System

The Smart Online Shopping System must fulfill a set of general requirements to support the functionality expected from a modern e-commerce platform.

Product Discovery and Catalog Management

- The system must allow customers to efficiently discover products through search tools, filtering options, and category navigation.
- Product pages must display detailed product information including descriptions, images, price, availability, ratings, and reviews.

- The system must also support product comparison features so users can evaluate multiple products before purchasing.

Shopping and Order Management

- The platform must provide customers with a shopping cart where products can be added, modified, or removed prior to purchase.
- The system must support secure checkout and order placement, while generating invoices or receipts for completed purchases.
- Customers must also be able to track order status and receive notifications regarding order confirmation, shipping updates, and delivery progress.

Payment and Transaction Processing

- The system must support secure payment transactions using multiple payment methods such as credit cards and digital wallets.
- Payment processing must ensure secure handling of financial data and confirmation of completed purchases.

Product Feedback and Recommendation

- The system must allow customers to submit product reviews and ratings based on their purchases.
- These reviews must be visible to other customers to help inform purchasing decisions.
- The platform must also generate personalized product recommendations based on browsing history, purchase behavior, and ratings.

Seller Management and Inventory Control

- The system must allow sellers to manage product listings and inventory levels.
- Sellers must be able to monitor stock availability, update product information, and receive alerts when inventory reaches low-stock thresholds.
- Predictive analytics tools should support sellers in forecasting inventory demand and optimizing product pricing.

Order Tracking and Logistics Integration

- The platform must allow customers to track order deliveries and view estimated delivery times.
- The system must synchronize order information with external logistics providers in order to update delivery status and provide real-time shipment information.

Notification and Alert System

The system must provide notifications to users regarding important platform activities, including:

- order confirmations
- shipping and delivery updates
- price changes
- promotional offers
- group buying opportunities

These notifications help ensure that users remain informed about their transactions and relevant platform events.

Sustainability and Transparency Features

- The system must provide transparency regarding the environmental impact of shipping options by displaying the estimated carbon footprint associated with delivery methods.
- This feature allows customers to make more environmentally conscious purchasing decisions.